

EXPLORER

DOCKER CAPSTON...

backend

.env

database_models.py

database.py

Dockerfile

main.py

models.py

requirements.txt

frontend

build

public

src

.env

Dockerfile

Dockerfile-non-root-user

nginx.conf

package-lock.json

package.json

.dockerignore

.gitignore

db.env

docker-compose.yaml

docker-compose.yaml

docker-compose.yaml > {} services

Run All Services | docker-compose.yml - The Compose specification establishes a standard for the definition of multi-container platform-agnostic applications (compose-)

1 services:

2 db:

3 image: postgres:16-alpine

4 env_file:

5 | - ./db.env

6 ports:

7 | - "5432:5432"

8 volumes:

9 | - pgdata:/var/lib/postgresql/data

10 healthcheck:

11 test: ["CMD-SHELL", "pg_isready -U postgres"]

12 interval: 5s

13 timeout: 3s

14 retries: 20

15

16 backend:

17 build:

18 | context: ./backend

19 env_file:

20 | - ./backend/.env

21 depends_on:

22 | db:

23 | | condition: service_healthy

24 ports:

25 | - "8000:8000"

26

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

powershell

+ v

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⋮

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PS C:\Users\jagad\OneDrive\Desktop\DevOps demo files\docker capstone project-1>

OUTLINE

TIMELINE

SESSI...

Locate provider.tf

4 wks ago

Tip: Use /create-agent to scaffold a custom agent for your workflow.

✕ docker-compose.y

Explore and underst...

+ ? 🗑 ↩

Local

Ln 1, Col 1

Spaces: 2

UTF-8

LF

{ Compose

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🔔

[Details](#)
[Status and alarms](#)
[Monitoring](#)
[Security](#)
[Networking](#)
[Storage](#)
[Tags](#)

Instance ID i-074cf65b730033c4e	Public IPv4 address 13.233.246.125 open address ↗	Private IPv4 addresses 172.31.10.87
IPv6 address —	Instance state Running	Public DNS ec2-13-233-246-125.ap-south-1.compute.amazonaws.com open address ↗

jagad@SURINENI MINGW64 ~/OneDrive/Desktop/DevOps demo files

\$ ssh -i docker-capstone-project-1.pem ec2-user@13.233.246.125

** WARNING: connection is not using a post-quantum key exchange algorithm.

** This session may be vulnerable to "store now, decrypt later" attacks.

** The server may need to be upgraded. See <https://openssh.com/pq.html>

~_#_ Amazon Linux 2023
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<https://aws.amazon.com/linux/amazon-linux-2023>

Last login: Tue Jul 28 13:29:23 2026 from 103.208.230.38

[ec2-user@ip-172-31-10-87 ~]\$ cd te-5th_and_6th_batch_assignments/surinenijagadeesh/"Docker capstone project-1"

[ec2-user@ip-172-31-10-87 Docker capstone project-1]\$ ls

```
[ec2-user@ip-172-31-10-87 ~]$ sudo systemctl start
```

```
[ec2-user@ip-172-31-10-87 ~]$ docker --version
```

```
Docker version 25.0.14, build 0bab007
```

```
[ec2-user@iip-172-31-10-87 ~]$ docker compose version  
Docker Compose version v2.39.1
```

backend db:env docker-compose.yml frontend

```
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ ls -la
```

```
total 32
```

```
drwxr-xr-x.  4 ec2-user ec2-user    117 Jul 28 13:16 .
drwxr-xr-x. 16 ec2-user ec2-user  16384 Jul 28 13:16 ..
-rw-r--r--.  1 ec2-user ec2-user    63 Jul 28 13:16 .dockerignore
-rw-r--r--.  1 ec2-user ec2-user    71 Jul 28 13:16 .gitignore
drwxr-xr-x.  2 ec2-user ec2-user    125 Jul 28 13:16 backend
-rw-r--r--.  1 ec2-user ec2-user    69 Jul 28 13:16 db.env
-rw-r--r--.  1 ec2-user ec2-user   592 Jul 28 13:16 docker-compose.yaml
drwxr-xr-x.  4 ec2-user ec2-user    144 Jul 28 13:16 frontend
```

```
drwxr-xr-x.  4 ec2-user ec2-user   144 Jul 28 13:16 frontend
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ cat .gitignore
.env
backend/.env
frontend/.env
node_modules/
build/
__pycache__/
```



```
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ docker compose up --build -d
```

```
[+] Running 12/12
```

```
✓ db Pulled 8.8s
  ✓ 55afa1ecc21d Pull complete 0.8s
  ✓ 0749aef11dca Pull complete 0.8s
  ✓ 2d3064b8615b Pull complete 0.8s
  ✓ ef6ab46d98b2 Pull complete 1.2s
  ✓ 689a434e3a86 Pull complete 1.2s
  ✓ 2f537278bc74 Pull complete 5.8s
  ✓ 4096c457dc0d Pull complete 5.8s
  ✓ c86ab38aa6db Pull complete 5.9s
  ✓ 713473b5b92b Pull complete 5.9s
  ✓ 7af924000309 Pull complete 5.9s
  ✓ cc7fa208b8a7 Pull complete 5.9s
```

```
[+] Running 7/7
```

```
✓ dockercapstoneproject-1-frontend Built
✓ dockercapstoneproject-1-backend Built
✓ Network dockercapstoneproject-1_default Created
✓ Volume "dockercapstoneproject-1_pgdata" Created
✓ Container dockercapstoneproject-1-db-1 Healthy
✓ Container dockercapstoneproject-1-backend-1 Started
✓ Container dockercapstoneproject-1-frontend-1 Started
```

```
dockercapstoneproject-1-frontend-1
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ docker ps
CONTAINER ID        IMAGE                               COMMAND                  CREATED            STATUS              PORTS                               NAMES
c8fb82353a43       dockercapstoneproject-1-frontend  "/docker-entrypoint..." 30 seconds ago    Up 22 seconds      0.0.0.0:80->80/tcp, :::80->80/tcp   dockercapstoneproject-1-frontend-1
a19434714784       dockercapstoneproject-1-backend   "uvicorn main:app --..." 30 seconds ago    Up 23 seconds      0.0.0.0:8000->8000/tcp, :::8000->8000/tcp  dockercapstoneproject-1-backend-1
700aa7aebec6       postgres:16-alpine               "docker-entrypoint.s..." 30 seconds ago    Up 29 seconds (healthy) 0.0.0.0:5432->5432/tcp, :::5432->5432/tcp  dockercapstoneproject-1-db-1
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ curl http://localhost:8000/products/
[{"name":"Phone","id":1,"price":699.99,"quantity":50,"description":"A smartphone"}, {"name":"Laptop","id":2,"price":999.99,"quantity":30,"description":"A powerful laptop"}, {"name":"Pen","id":3,"price":1.99,"quan
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ Read from remote host 13.233.246.125: Connection reset by peer
```

```
INFO: 172.18.0.1:47558 - GET /products/ HTTP/1.1 200 OK
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ curl http://localhost:8000/products/
[{"name":"Phone","id":1,"price":699.99,"quantity":50,"description":"A smartphone"}, {"name":"Laptop","id":2,"price":999.99,"quantity":30,"description":"A powerful laptop"}, {"name":"Pen","id":3,"price":1.99,"quantity":100,"description":"A blue ink pen"}, {"name":"Table","id":4,"price":199.99,"quantity":20,"description":"A wooden table"}]
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ curl http://13.233.246.125:8000
curl http://13.233.246.125:8000/products/
```

backend db.env docker-compose.yaml frontend

```
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ curl http://13.233.246.125:8000/products/  
[{"name":"Phone","id":1,"price":699.99,"quantity":50,"description":"A smartphone"}, {"name":"Laptop","id":2,"price":999.99,"quantity":30,"description":"A powerful laptop"}, {"name":"Pen","id":3,"price":1.99,"quantity":100,"description":"A blue ink pen"}, {"name":"Table","id":4,"price":199.99,"quantity":20,"description":"A wooden table"}]  
[ec2-user@ip-172-31-10-87 Docker capstone project-1]$ Read from remote host 13.233.2
```

Add Product

ID	Name	Description	Price
Quantity			
Add			

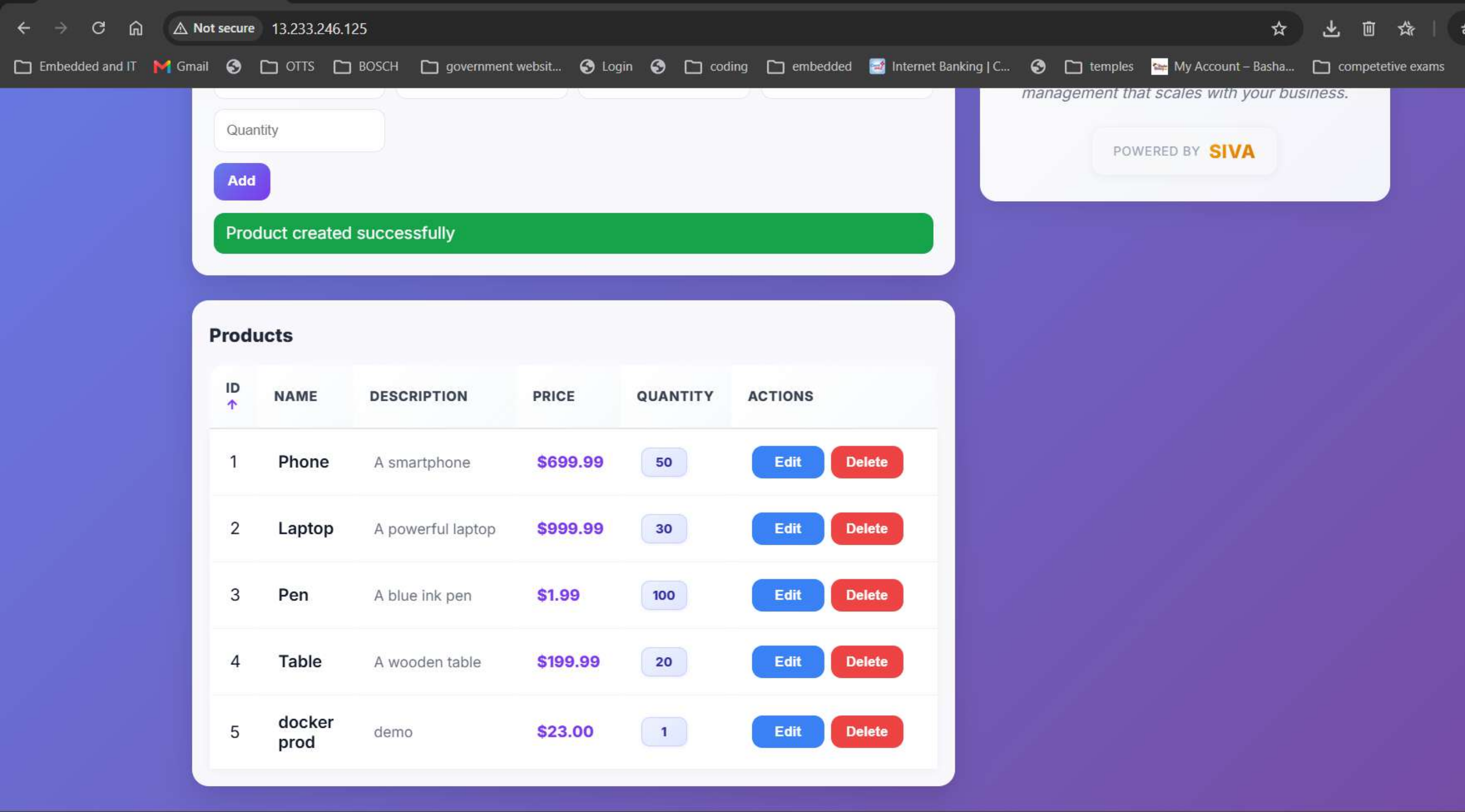
Track. Manage. Grow.

Streamline your inventory with smart product management that scales with your business.

POWERED BY SIVA

Products

ID ↑	NAME	DESCRIPTION	PRICE	QUANTITY	ACTIONS
1	Phone	A smartphone	\$699.99	50	Edit Delete
2	Laptop	A powerful laptop	\$999.99	30	Edit Delete
3	Pen	A blue ink pen	\$1.99	100	Edit Delete
4	Table	A wooden table	\$199.99	20	Edit Delete



Quantity

Add

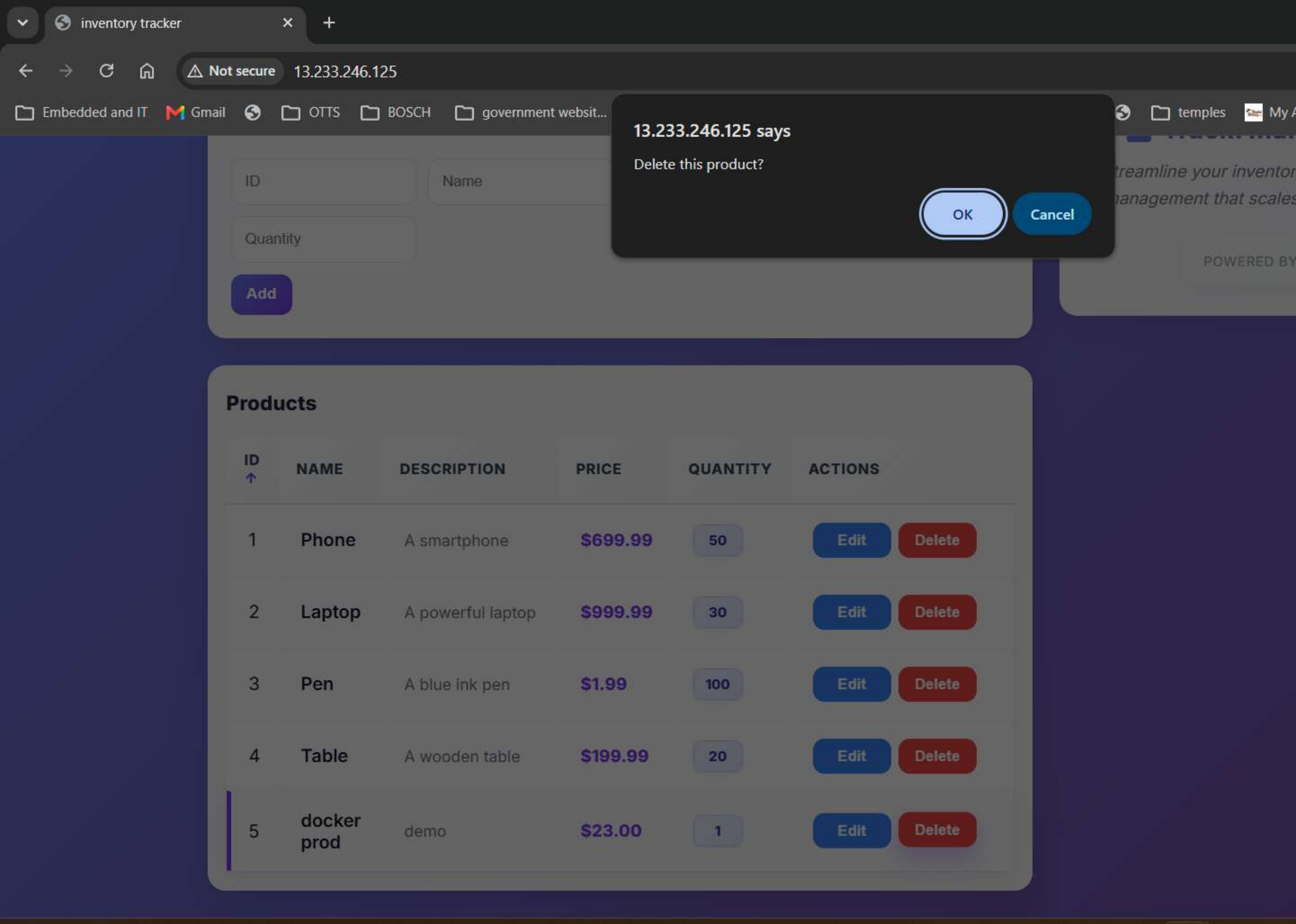
Product created successfully

management that scales with your business.

POWERED BY SIVA

Products

ID ↑	NAME	DESCRIPTION	PRICE	QUANTITY	ACTIONS
1	Phone	A smartphone	\$699.99	50	<button>Edit</button> <button>Delete</button>
2	Laptop	A powerful laptop	\$999.99	30	<button>Edit</button> <button>Delete</button>
3	Pen	A blue ink pen	\$1.99	100	<button>Edit</button> <button>Delete</button>
4	Table	A wooden table	\$199.99	20	<button>Edit</button> <button>Delete</button>
5	docker prod	demo	\$23.00	1	<button>Edit</button> <button>Delete</button>



13.233.246.125 says

Delete this product?

OK

Cancel

Products

ID ↑	NAME	DESCRIPTION	PRICE	QUANTITY	ACTIONS
1	Phone	A smartphone	\$699.99	50	<button>Edit</button> <button>Delete</button>
2	Laptop	A powerful laptop	\$999.99	30	<button>Edit</button> <button>Delete</button>
3	Pen	A blue ink pen	\$1.99	100	<button>Edit</button> <button>Delete</button>
4	Table	A wooden table	\$199.99	20	<button>Edit</button> <button>Delete</button>
5	docker prod	demo	\$23.00	1	<button>Edit</button> <button>Delete</button>

Add Product

Add

Product deleted successfully

Products

ID ↑	NAME	DESCRIPTION	PRICE	QUANTITY	ACTIONS
1	Phone	A smartphone	\$699.99	50	EditDelete
2	Laptop	A powerful laptop	\$999.99	30	EditDelete
3	Pen	A blue ink pen	\$1.99	100	EditDelete
4	Table	A wooden table	\$199.99	20	EditDelete